4.9 Modeling with Polynomial Functions

Name _____ Period ____ Date _____

Algebra II 4.9 (30-minute plan) · Student sheet

I can…

- write cubic functions given the x-intercepts and a point.
- use finite differences to find the degree of a polynomial model.
- use technology to find polynomial models for real-life data.

Vocabulary · fill in the missing word

Finite differences — differences of _____ y-values when the x-values are equally spaced

Degree from differences — if the nth differences are _____ (and nonzero), the data fit a polynomial of degree n

Regression — _____ that finds the polynomial of a given degree that best fits data

Part 1 · Notes and examples (fill in each note, then work the example)

Cubic from x-intercepts: Cubic from x-intercepts p, q, r: f(x) = _____. Use another point to find _____.

Example 1

Write the cubic function whose graph has x-intercepts −2, 1, and 3 and passes through (0, −6).

Finite differences: Constant nth differences → degree _____.

Example 2

Use finite differences to find the degree of the polynomial function that fits the data. Then find the function.

x	1	2	3	4	5	6
f(x)	2	9	28	65	126	217

Part 2 · Partner practice

Partner practice

Find the degree of the polynomial that fits the data: x = 1, 2, 3, 4, 5; y = 3, 6, 11, 18, 27.